Pranav BJ

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EDUCATION

Acharya Institute of Technology | *Computer Science Engineering and Data Science* *Dec 2022 - Jun 2026*

- **Cumulative GPA: 8.37/10.00** *(As of June 2025)*
- **Coursework:** Big Data Analytics, Artificial Intelligence and Machine Learning, Software Engineering, Cyber Security, Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, DevOps.

EXPERIENCE

Front-End Developer Intern | **Accuracy.ai** | *Next.js, Tailwind CSS, TypeScript, Git, GitHub, Figma* *Jun 2025 – Aug 2025*

- Integrated REST APIs to fetch, manage, and render dynamic data across multiple application modules, enabling seamless frontend–backend interactions.
- Built **15+ reusable responsive UI components** using Next.js and Tailwind CSS, improving development efficiency and consistency across the application.
- Translated Figma designs into **pixel-perfect production-ready interfaces** in an Agile/Scrum environment, reducing feature development time by **25%**

PROJECTS

AI Stock ETL Pipeline with Data Quality Monitoring | [Github] | *Python, Apache Airflow, MySQL, Docker, Lang Chain, Llama 3.1*

- Built a Dockerized Airflow ETL pipeline extracting livestock data from Rapid API and loading ~5,000 records/day (~150K/month) into MySQL with validation, deduplication, and retry logic achieving 99.5% data integrity.
- Designed scalable Airflow DAG workflows using XCom communication and environment-based secret management for reliable, fault-tolerant pipeline execution.
- Integrated a Lang Chain + Llama 3.1 (Groq) AI agent to detect price anomalies, missing values, and volume spikes after each pipeline run for automated data quality monitoring.

Multimodal Synthetic Healthcare Generator | [Github] | *React.js, Python, Tailwind CSS, Flask, FastAPI, Docker, MongoDB, Render, FLAN-T5*

- Designed– and secured RESTful microservice APIs across Flask and FastAPI with API-key authentication, applying separation of concerns and service isolation principles generating synthetic healthcare datasets at 100–10,000 records per run backed by MongoDB.
- Integrated FLAN-T5 (Hugging Face Transformers) to auto-generate natural-language summaries of dataset statistics, delivering structured insights (~227 characters) in ~9s per CPU inference call.
- Containerized all services using Docker and deployed on Render and built a React.js interface for configurable generation, CSV export, and time-series visualizations owning the full stack from API design to cloud deployment.

Distributed Multi-Currency Transaction Engine | [Github] | *Java 17, Spring Boot 3, Spring Data JPA, H2, JUnit 5, Mockito, Docker*

- Built a production-style multi-currency REST API in Java 17 + Spring Boot with live exchange rate integration supporting cross-border fund transfers.
- Implemented Strategy & Repository design patterns across a 3-layer architecture (Controller → Service → Repository) following OOAD principles.
- Achieved 85%+ test coverage using JUnit 5 and Mockito with mocks and stubs, containerized the application using Docker for portable deployment

SKILLS

Languages: C++, Java, Python, JavaScript, Typescript, SQL
Backend: Node.js, Express.js, Flask, RESTful API Design, Spring Boot
Databases: MySQL, PostgreSQL, MongoDB
Cloud & DevOps: Docker, Apache Airflow, Git, CI/CD (GitHub Actions), Linux, NPM, AWS, Render, Vercel
Frontend: React.js, Next.js, Tailwind CSS, Redux, Vite

ACHIEVEMENTS

- Selected for incubation by the Institute's Entrepreneurship Cell for presenting an innovative solution during the campus Ideathon.
- Selected at the Institute level for Smart India Hackathon (SIH) 2024 and participated in multiple regional hackathons.
- Solved 270+ problems across LeetCode and Striver's SDE Sheet covering Arrays, DP, Graphs, Trees, and Sliding Window.